Beyond-Diagonal RIS-aided Outdoor-to-Indoor mmWave Communications

Hesam Nezhadmohammad, Reza Asvadi, *Senior Member, IEEE*, and Abbas Pirhadi, *Member, IEEE*

***Abstract*—Outdoor-to-indoor millimeter-wave (mmWave) communications are hindered by high penetration losses and unpredictable blockages, which disrupt line-of-sight connectivity between base station (BS) and indoor users. To address these challenges, this paper investigates the application of beyond-diagonal reconfigurable intelligent surface (BD-RIS) for assisting outdoor-to-indoor mmWave communications. Unlike conventional RIS, BD-RIS is a novel and more general framework whose scattering matrix is no longer restricted to being diagonal, thereby enabling more effective manipulation and enhanced control of waves than conventional designs. We propose an alternating optimization framework that jointly optimizes BS beamforming and the BD-RIS scattering matrix to maximize the weighted sum-rate (WSR) of all users. The framework applies to fully-, group-, and single-connected BD-RIS architectures, and incorporates symmetric unitary constraints of BD-RIS and a probabilistic blockage model. Simulations show that the proposed design achieves nearly 48% higher WSR compared to the conventional RIS design and delivers up to 99% improvement with an increasing number of RIS elements under mild blockage conditions. These results demonstrate that BD-RIS offers a robust and efficient solution for enhancing outdoor-to-indoor mmWave communications even in the presence of blockage.**

***Index Terms*—Beyond diagonal reconfigurable intelligent surface (BD-RIS), millimeter-wave (mmWave), outdoor-to-indoor communication, weighted sum-rate (WSR) maximization.**

I. Introduction

A. Background

As mobile users spend nearly 87% of their time indoors [1], ensuring robust indoor wireless coverage has become a pressing challenge for next-generation networks. Although indoor small-cell base stations (BSs) is a conventional solution, they involve high deployment costs, complex backhaul, and potential inter-cell interference in dense urban areas. As a result, there is increasing interest in alternative strategies that can deliver high-speed wireless service to indoor users directly from outdoor infrastructure without relying on indoor active equipment.

Outdoor-to-indoor millimeter-wave (mmWave) communication has emerged as a practical solution for providing high-speed wireless service to indoor users by leveraging existing outdoor base stations [2], [3]. It benefits from the large bandwidth available at mmWave frequencies, enabling multi-gigabit data rates [4], [5]. However, mmWave signals encounter major propagation challenges, such as high free-space path loss and limited ability to pass through building materials. For example, walls and tinted glass can cause penetration losses of over 28–40 dB [2], [6], especially when the base station is not aligned with a window or transparent surface. In addition to these fixed losses, mmWave links are easily blocked by moving objects like pedestrians and vehicles, as well as static obstacles such as trees or buildings [4], [7], [8]. These blockages often interrupt the line-of-sight (LoS) connection and cause sudden signal degradation or even complete outage. A single person, for instance, can block the main path for several hundred milliseconds [4], leading to noticeable service interruptions. In dense areas, the combined effect of penetration loss and random blockages significantly weakens link reliability. This motivates the use of auxiliary techniques to maintain stable connectivity despite these impairments.

One promising approach to improve link reliability under such impairments is deployment of reconfigurable intelligent surfaces (RISs), which serve as passive and energy-efficient solutions to enhance signal propagation. Comprising nearly passive elements that can adjust the phase of incident waves, RISs are capable of reconfiguring the wireless channel by creating virtual LoS paths, thereby mitigating blockage and extending mmWave coverage [9]–[11]. Recent field trials in commercial 5G frequency range 2 networks above 24 GHz have shown that RIS can substantially improve indoor coverage, with throughput gains of up to 227.69% [12], [13]. Despite these promising gains, conventional RISs rely on diagonal phase-shift matrices which limits their ability to support outdoor-to-indoor links that require signal transmission through walls or windows [6], [14].

To address these limitations, a recent advance in RIS known as beyond-diagonal reconfigurable intelligent surface (BD-RIS) has been introduced [15]. By allowing tunable off-diagonal entries in the scattering matrix, which were previously restricted to zero in conventional RISs, BD-RIS enables more flexible wave manipulation and enhanced control over signal propagation [15]–[17]. BD-RIS can be implemented using fully-connected, group-connected, single-connected, and more sophisticated architectures,[1] providing a wide range of trade-offs between hardware complexity and performance [15], [16]. Notably, BD-RIS supports reflective, transmissive, and hybrid modes [17], thereby enabling advanced signal control strategies essential for challenging scenarios like outdoor-to-indoor communication where conventional RISs often fall short [13].

H. Nezhadmohammad, R. Asvadi, and A. Pirhadi are with the Department of Telecommunications, Faculty of Electrical Engineering, Shahid Beheshti University, Tehran 1983963113, Iran (email: h.nezhadmohammad@mail.sbu.ac. ir; {r.asvadi, a.pirhadi}@sbu.ac. ir).

[1]Recent works have proposed additional architectures such as tree-connected and forest-connected [18], band-connected and stem-connected [19], and dynamically connected BD-RIS [20]. We refer the reader to the tutorial in [13] for a comprehensive overview of BD-RIS modeling, architectures, and applications.

B. Prior Works

Several recent works have investigated the application of conventional RIS and related architectures to enhance outdoor-to-indoor mmWave communication. The authors in [21] proposed intelligent omni-surfaces (IOSs) mounted on building exteriors to refract mmWave signals from an outdoor base station to indoor users, jointly optimizing their placement and phase shifts to maximize energy efficiency. In [22], a gradient descent algorithm was developed to optimize the positioning and orientation of passive RISs to expand indoor coverage in non-line-of-sight (NLoS) regions. The study in [23] considered a cascaded RIS setup, where outdoor and indoor RISs sequentially refracted the mmWave signal toward the user, with joint active and passive beamforming to enhance the received SNR. The work in [6] integrated a refracting RIS into the building wall and demonstrated improved coverage probability compared to a relay-aided baseline. Finally, [14] investigated an active RIS-assisted mmWave system, where the authors optimized the BS and RIS precoders to enhance the downlink data rate for indoor users.

Despite these contributions, several important limitations remain unaddressed across these studies. First, none of [22], [23], or [6] accounted for multi-user interference, which is critical in dense indoor deployments. Second, [21], [6], and [14] relied on RIS or IOS architectures modeled via diagonal phase-shift matrices, which cannot capture group-connected or fully-connected behavior inherent to BD-RIS designs [17]. Third, dynamic or static blockage effects were not incorporated in [22], [23], or [14], limiting the practical validity of their channel assumptions. Lastly, while [14] employed active RIS to overcome passive limitations, such designs introduce hardware complexity, higher power consumption, and potential signal degradation due to amplified noise, all of which can be effectively addressed by BD-RIS through its passive inter-element control mechanism.

Several studies have demonstrated the benefits of BD-RIS across various wireless communication scenarios. By enabling joint control of phase and amplitude through inter-element connections [16], [17], [24], BD-RIS improves overall performance by achieving higher spectral efficiency and signal strength than conventional RIS, in wideband transmission [25], full-duplex systems [26], and aerial platforms [27]. Recent works have also explored transmitter-side BD-RIS architectures for multi-user integrated sensing and communications, showing improved performance and reduced hardware complexity using fully-connected scattering matrix of BD-RIS [28], [29].

C. Motivation and Contributions

While there has been notable progress, two key gaps remain in the literature. First, the application of BD-RIS for outdoor-to-indoor mmWave communication has not been studied. However, its support for transmissive operation and inter-element connections makes it especially suitable for such environments by enabling directional signal flow between elements [17], [30]. This is further motivated by a recent study that highlights BD-RIS as a promising solution for improving coverage in both indoor and outdoor scenarios [13]. Second, there is no unified and efficient framework for evaluating different BD-RIS architectures under realistic propagation conditions, including building penetration loss and random blockages. This work addresses these gaps with the following main contributions:

- We develop a system model for BD-RIS-assisted outdoor-to-indoor mmWave communication in a multi-user multiple-input single-output (MU-MISO) scenario, incorporating key propagation impairments such as penetration loss through building materials and random link blockages using a probabilistic blockage model. Based on this model, we formulate a weighted sum-rate (WSR) maximization problem under a transmit power constraint at the BS and symmetric and unitary constraints on the BD-RIS scattering matrix.
- We propose an alternating optimization (AO) framework that jointly updates the BS transmit beamforming vectors and the BD-RIS scattering matrix. The BS update employs the weighted minimum mean square error (WMMSE) method, while the BD-RIS update is performed via manifold optimization. The algorithm supports fully-connected, group-connected, and single-connected architectures, enabling a unified evaluation across different BD-RIS designs.
- We conduct a comparative analysis of fully-, group-, and single-connected BD-RIS architectures under varying numbers of RIS elements, users, blockage densities, and power budgets at the BS. The evaluation highlights the trade-offs between these architectures in terms of achievable WSR, outage probability, and robustness to random link blockages.
- Simulation results validate the effectiveness of the proposed AO algorithm, demonstrating near-optimal WSR performance with significantly lower runtime compared to convex solvers. Among the considered architectures, the fully-connected BD-RIS consistently yields the highest WSR and lowest outage probability across varying blockage densities, while the group-connected and single-connected architectures exhibit lower performance due to their simpler structural designs. These findings provide practical insights into the trade-offs involved in selecting BD-RIS architectures for outdoor-to-indoor mmWave systems.

This study represents one of the first investigations of using BD-RIS for outdoor-to-indoor mmWave systems and provides a unified performance evaluation across BD-RIS architectures under realistic channel conditions.

D. Organization and Summary of Notations

The rest of the paper is organized as follows. Section II presents the system, channel, and blockage models, along with the problem formulation. Section III describes the proposed optimization framework. Simulation results are provided in Section IV, and Section V concludes the paper. Table I summarizes the notational conventions used throughout the paper.

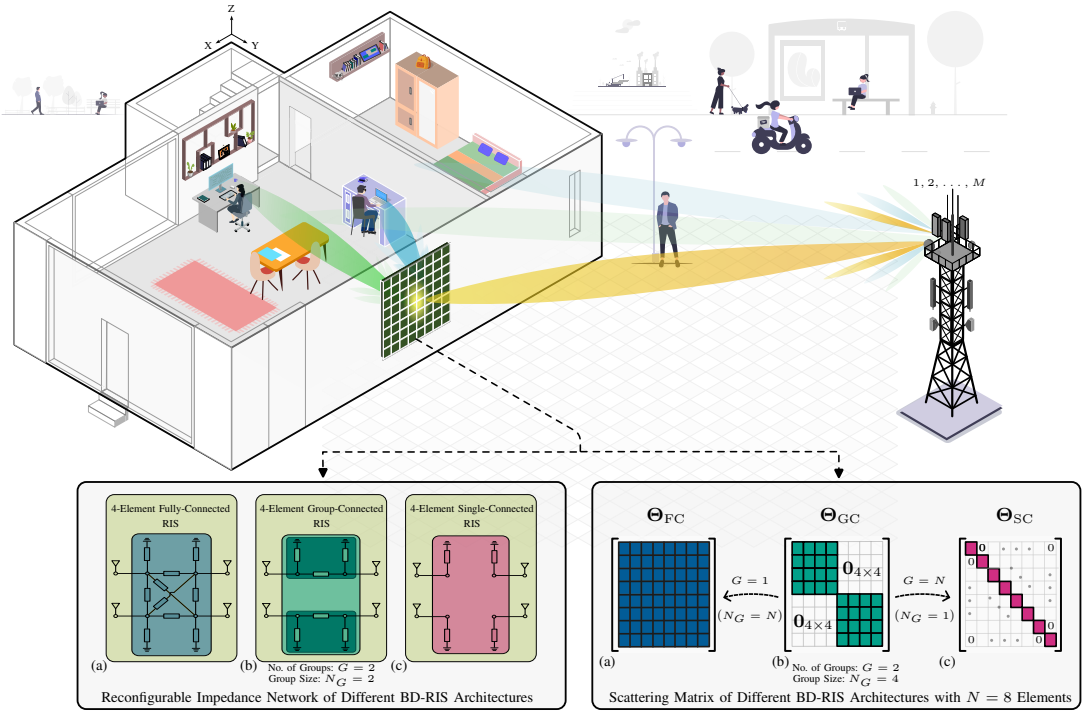


Fig. 1. System model of BD-RIS-assisted outdoor-to-indoor mmWave communication with different BD-RIS architectures: (a) fully-connected, (b) group-connected, (c) single-connected.

TABLE I
SUMMARY OF THE NOTATIONS

Notation	Description
$\mathbf{a}$	Bold lowercase letters denote vectors
$\mathbf{A}$	Bold uppercase letters denote matrices
$\mathbb{C}, \mathbb{R}$	Sets of complex and real numbers
j	Imaginary unit, $j = \sqrt{-1}$
$(\cdot)^*$	Complex conjugate operator
$(\cdot)^T$	Transpose operator
$(\cdot)^H$	Conjugate transpose operator
$(\cdot)^{-1}$	Matrix inverse
$\Re(\cdot)$	Real part of a complex number
$\angle x$	Phase angle of a complex scalar x
$\lvert a \rvert$	Absolute value of scalar a
$\lVert \mathbf{A} \rVert_F$	Frobenius norm of matrix $\mathbf{A}$
$\mathbf{I}_L$	Identity matrix of size $L \times L$
$\mathbf{0}$	Zero matrix of compatible size
$\mathsf{Tr}(\cdot)$	Trace of a matrix
$\mathbb{E}[\cdot]$	Statistical expectation
$\mathbf{A} \odot \mathbf{B}$	Hadamard product of matrices
$\mathbf{A} \otimes \mathbf{B}$	Kronecker product of matrices
$\mathsf{diag}(\cdot)$	Diagonal matrix with specified entries
$\mathsf{blkdiag}(\cdot)$	Constructs a block matrix with input matrices on the diagonal and zeros elsewhere

II. System Model and Problem Formulation

We consider MU-MISO downlink communication system empowered by a BD-RIS operating in transmissive mode [17]. The setup consists of a base station (BS) equipped with a uniform linear array (ULA) of M antennas, a BD-RIS equipped with uniform rectangular array (URA) with N transmissive elements, and K single-antenna users indexed by $\mathcal{K} = \{1, 2, \ldots, K\}$. The BD-RIS elements are indexed by $\mathcal{N} = \{1, 2, \ldots, N\}$ and configured using a smart controller that also facilitates information exchange with the BS.

In the envisioned outdoor-to-indoor mmWave communication scenario, the BS transmits signals to indoor users through the BD-RIS. By leveraging the transmissive properties of the BD-RIS, the incident signals pass through the surface and are refracted into the indoor environment. Fig. 1 illustrates the system architecture, including the BS, BD-RIS, and users.

The transmitted signal from the BS reaches the users through both the direct BS-user link and the BD-RIS-assisted link. To characterize the received signal, we first define the equivalent channel for user k as $\mathbf{c}_k \triangleq (\mathbf{h}_k^H \mathbf{\Theta} \mathbf{H} + \mathbf{d}_k^H)^H$, where $\mathbf{H} \in \mathbb{C}^{N \times M}$, $\mathbf{h}_k \in \mathbb{C}^{N \times 1}$, and $\mathbf{d}_k \in \mathbb{C}^{M \times 1}$ denote the BS-RIS, RIS-user k, and BS-user k mmWave channels, respectively, $\forall k \in \mathcal{K}$ and matrix $\mathbf{\Theta} \in \mathbb{C}^{N \times N}$ represents the BD-RIS scattering matrix. Let $\mathbf{s} \triangleq [s_1, \ldots, s_K]^T \in \mathbb{C}^{K \times 1}$ denote the data stream vector for the users, where $\mathbb{E}[\mathbf{s}\mathbf{s}^H] = \mathbf{I}_K$, and let $\mathbf{W} \triangleq [\mathbf{w}_1, \ldots, \mathbf{w}_K] \in \mathbb{C}^{M \times K}$ be the corresponding beamforming matrix at the BS, where $\mathbf{w}_k \in \mathbb{C}^{M \times 1}$ is the linear precoder for user k, $\forall k \in \mathcal{K}$. The transmitted signal at the BS is $\mathbf{x} = \sum_{j \in \mathcal{K}} \mathbf{w}_j s_j \in \mathbb{C}^{M \times 1}$, and the transmit power constraint is $\lVert \mathbf{W} \rVert_F^2 \leq P_t$, where P_t denotes the maximum

transmit power at the BS.

The signal received at user k can now be expressed as

$$\begin{aligned} y_k &= \underbrace{\mathbf{h}_k^H \boldsymbol{\Theta} \mathbf{H} \mathbf{x}}_{\text{BD-RIS-aided link}} + \underbrace{\mathbf{d}_k^H \mathbf{x}}_{\text{Direct link}} + n_k \\ &= \underbrace{(\mathbf{h}_k^H \boldsymbol{\Theta} \mathbf{H} + \mathbf{d}_k^H)}_{\triangleq \mathbf{c}_k^H \in \mathbb{C}^{1\times M}} \sum_{j\in\mathcal{K}} \mathbf{w}_j s_j + n_k \\ &= \mathbf{c}_k^H \mathbf{w}_k s_k + \mathbf{c}_k^H \sum_{j\in\mathcal{K}, j\neq k} \mathbf{w}_j s_j + n_k, \ \forall k \in \mathcal{K}, \end{aligned} \tag{1}$$

where $n_k \sim \mathcal{CN}(0, \sigma_k^2)$ is the additive white Gaussian noise with zero mean and variance σ_k^2.

A. Channel Model

The channel between the BS and user k is generated according to the widely used Saleh-Valenzuela (SV) channel model [7], which is expressed as

$$\mathbf{d}_k = \sqrt{\frac{M}{L_{\text{BU}}}} \sum_{p=1}^{L_{\text{BU}}} \alpha_p \mathbf{a}_{\text{ULA}}(\phi_p), \ \forall k \in \mathcal{K}, \tag{2}$$

where L_{BU} is the number of NLoS paths, α_p is the complex gain associated with the p-th path, ϕ_p is the associated angle of departure (AoD), $\mathbf{a}_{\text{ULA}}(\phi_p) \in \mathbb{C}^{M\times 1}$ represents the normalized array response vector of the BS. The channels between the RIS and user k, and between the BS and the RIS, are generated according to the aforementioned SV model under LoS scenarios. Specifically, the channel between the RIS and user k is modeled as a sum of LoS and NLoS components using the SV model as

$$\begin{aligned} \mathbf{h}_k = &\sqrt{\frac{N}{L_{\text{RU}}}} \Bigg(\underbrace{\varrho_{0,k} \mathbf{a}_{\text{URA}}(\nu_{0,k}, \vartheta_{0,k})}_{\mathbf{h}_k^{\text{LoS}}} \\ &+ \underbrace{\sum_{l=1}^{L_{\text{RU}}-1} \varrho_{l,k} \mathbf{a}_{\text{URA}}(\nu_{l,k}, \vartheta_{l,k})}_{\mathbf{h}_k^{\text{NLoS}}} \Bigg), \ \forall k \in \mathcal{K}, \end{aligned} \tag{3}$$

where L_{RU} is the number of paths between the RIS and user k. The term $\varrho_{l,k}$ represents the complex path gain of the l-th path for user k, with $l = 0$ corresponding to the LoS component. The variables $\nu_{l,k}$ and $\vartheta_{l,k}$ denote the azimuth and elevation AoD, respectively, associated with the l-th path. The vector $\mathbf{a}_{\text{URA}}(\nu, \vartheta) \in \mathbb{C}^{N\times 1}$ is the normalized array response of $U \times V$ RIS with respect to the two-dimensional angles (ν, ϑ), where U and V denote the number of RIS elements along the horizontal axis and vertical axis, respectively. On the other hand, the channel between the BS and the RIS is characterized by the SV channel model given as

$$\begin{aligned} \mathbf{H} = &\sqrt{\frac{MN}{L_{\text{BR}}}} \Bigg(\underbrace{\alpha_0 \mathbf{a}_{\text{URA}}(\nu_0, \vartheta_0) \mathbf{a}_{\text{ULA}}^H(\phi_0)}_{\mathbf{H}^{\text{LoS}}} \\ &+ \underbrace{\sum_{i=1}^{L_{\text{BR}}-1} \alpha_i \mathbf{a}_{\text{URA}}(\nu_i, \vartheta_i) \mathbf{a}_{\text{ULA}}^H(\phi_i)}_{\mathbf{H}^{\text{NLoS}}} \Bigg), \end{aligned} \tag{4}$$

where α_0 denotes the complex gain of the LoS component, ν_i (ϑ_i) denotes the azimuth (elevation) angles of arrival (AoA) associated with the i-th NLoS path. $\mathbf{a}_{\text{URA}}(\nu, \vartheta)$ and $\mathbf{a}_{\text{ULA}}(\phi)$ are defined as

$$\begin{aligned} \mathbf{a}_{\text{URA}}(\nu, \vartheta) = \Big[&1, \ldots, e^{\frac{j2\pi d}{\lambda}(u \sin(\nu)\sin(\vartheta) + v\cos(\vartheta))}, \ldots, \\ &e^{\frac{j2\pi d}{\lambda}((U-1)\sin(\nu)\sin(\vartheta) + (V-1)\cos(\vartheta))} \Big]^T, \\ \mathbf{a}_{\text{ULA}}(\phi) = \Big[&1, e^{\frac{-j2\pi d}{\lambda}\sin(\phi)}, \ldots, e^{\frac{-j2\pi d}{\lambda}(M-1)\sin(\phi)} \Big]^T. \end{aligned}$$

The array element spacing d of both ULA and URA is assumed to be $\lambda/2$, where λ denotes the signal wavelength.

B. Blockage Model

MmWave communication relies on the characteristics of the LoS path, as NLoS components typically exhibit 20–30 dB lower power levels than the dominant LoS component [4]. Field measurements in urban environments show that 20% to 60% of mmWave links experience blockage events, which can reduce the achievable rate by up to an order of magnitude [4], [31]. Consequently, accurate modeling of blockage effects is essential for conducting reliable performance analysis in mmWave communication systems. To address this, we adopt a probabilistic blockage model that captures the effects of blockage density and link distance, as proposed in [31], [32]. Let $d_{\text{RU},k}$ and η represent the distance between the RIS and user k, and the blockage density, respectively. Similar to [31], [32], the blockage between RIS and each user k is defined using a Bernoulli random variable with the parameter η and expressed as

$$\mathbf{h}_k^{\text{LoS}} = \begin{cases} \varrho_{0,k} \mathbf{a}_{\text{URA}}(\nu_{0,k}, \vartheta_{0,k}) & \text{w.p.} \quad e^{-\eta d_{\text{RU},k}} \\ \mathbf{0} & \text{w.p.} \quad 1 - e^{-\eta d_{\text{RU},k}} \end{cases} \tag{5}$$

where η depends on the density and the average size of obstacles blocking the dominant LoS path. The term $e^{-\eta d_{\text{RU},k}}$ represents the probability that an unobstructed LoS exists over distance $d_{\text{RU},k}$ [32]. A smaller value of η indicates fewer obstacles, thus increasing the LoS probability. For example, at 28 GHz, a single mobile blocker can obstruct the LoS path and cause 20–30 dB attenuation for several hundred milliseconds [33], effectively modeled as $\mathbf{h}_k^{\text{LoS}} = \mathbf{0}, \forall k \in \mathcal{K}$. We assume that blockage events are independent across user links and that NLoS multipath components remain present even if the LoS path is blocked. Additionally, we assume that the LoS component of the channel between the BS and the RIS is never obstructed, i.e., $\Pr\{\mathbf{H}^{\text{LoS}} = \mathbf{0}\} = 0$, and that all user links share the same blockage density, as in [11]. Since measurements show that the LoS power exceeds the sum of NLoS power by approximately 13 dB [4], [11], neglecting blockage effects can lead to overly optimistic performance estimates [8], [32]. Consequently, in this work, we use the parameter η to analyze the impact of varying blockage densities on system performance.

C. Beyond-Diagonal Reconfigurable Intelligent Surface Model

Unlike conventional RIS designs where the scattering matrix $\boldsymbol{\Theta}$ is restricted to a diagonal structure, BD-RIS introduces

tunable inter-element connections, enabling joint phase and amplitude control. This added structural flexibility enhances the surface's ability to manipulate incident waves and adapt to complex propagation environments. To balance performance and hardware complexity, three representative BD-RIS architectures have been studied in this work: fully-connected, group-connected, and single-connected.

1) Single-Connected: In this architecture, each element operates independently, resulting in a diagonal scattering matrix $\mathbf{\Theta}$, i.e.,

$$\mathbf{\Theta} = \mathsf{diag}(e^{j\phi_1}, e^{j\phi_2}, \ldots, e^{j\phi_N}), \tag{6}$$

where $\phi_n \in [0, 2\pi)$ denotes the phase shift of the n-th element. This architecture, also referred to as conventional (diagonal) RIS, has been extensively studied in wireless communication systems [10], [11]. While this design avoids inter-element connections and reduces circuit complexity [13], [15], it allows control over only N scattering parameters. This limitation restricts wave manipulation flexibility and degrades performance in complex environments.

2) Fully-Connected: In the fully-connected architecture, each element is connected to every other element through a reconfigurable impedance network [15]. The resulting scattering matrix $\mathbf{\Theta}$ is symmetric and unitary [13], [15], i.e.,

$$\mathbf{\Theta} = \mathbf{\Theta}^T, \ \mathbf{\Theta}^H\mathbf{\Theta} = \mathbf{I}_N. \tag{7}$$

This design generalizes D-RIS by allowing control over all $N(N+1)/2$ parameters in $\mathbf{\Theta}$, including both diagonal and off-diagonal elements [13], [15]. Such full inter-element connectivity significantly enhances wave manipulation flexibility, enabling superior performance in complex propagation environments.

3) Group-Connected: To integrate the advantages of both single- and fully-connected architectures, the group-connected design partitions the N elements into G disjoint groups. These groups are indexed by the set $\mathcal{G} = \{1, \cdots, G\}$, and each group consists of $N_G = \frac{N}{G}$ elements. Within each group, the elements are interconnected following the fully-connected structure, while no connections exist between elements in different groups. The resulting scattering matrix $\mathbf{\Theta}$ takes a block-diagonal form, i.e.,

$$\mathbf{\Theta} = \mathsf{diag}(\mathbf{\Theta}_1, \ldots, \mathbf{\Theta}_G), \ \mathbf{\Theta}_g = \mathbf{\Theta}_g^T, \ \mathbf{\Theta}_g^H\mathbf{\Theta}_g = \mathbf{I}_{N_G}, \forall g \in \mathcal{G}, \tag{8}$$

where $\mathbf{\Theta}_g$ denotes the symmetric and unitary scattering matrix corresponding to the g-th group. Notably, the single-connected and fully-connected architectures are special cases of the group-connected model, corresponding to $G = N$ and $G = 1$, respectively. This architecture offers a flexible trade-off between performance and circuit complexity [13], [15], making it a practical and scalable choice for real-world deployments.

D. Problem Formulation

The k-th user treats all signals from other users as interference. Accordingly, the signal-to-interference-plus-noise ratio (SINR) for decoding symbol s_k at user k is given by

$$\gamma_k = \frac{|\mathbf{c}_k^H\mathbf{w}_k|^2}{\sum_{j\in\mathcal{K}, j\neq k}|\mathbf{c}_k^H\mathbf{w}_j|^2 + \sigma_k^2}. \tag{9}$$

Our objective is to jointly optimize the BS beamforming matrix $\mathbf{W}$ and the BD-RIS scattering matrix $\mathbf{\Theta}$ to maximize the WSR across all users , while satisfying the total power constraint at the BS and structural constraints on $\mathbf{\Theta}$ as defined in (8). The resulting WSR maximization problem for the downlink BD-RIS-assisted outdoor-to-indoor mmWave system is formulated as

$$\max_{\mathbf{W},\mathbf{\Theta}} \quad f_{\mathrm{A}}(\mathbf{W}, \mathbf{\Theta}) = \sum_{k\in\mathcal{K}} \omega_k R_k \tag{10a}$$

$$\text{s.t.} \quad \|\mathbf{W}\|_F^2 \leq P_t, \tag{10b}$$

$$\mathbf{\Theta} = \mathsf{diag}(\mathbf{\Theta}_1, \ldots, \mathbf{\Theta}_G), \tag{10c}$$

$$\mathbf{\Theta}_g = \mathbf{\Theta}_g^T, \ \mathbf{\Theta}_g^H\mathbf{\Theta}_g = \mathbf{I}_{N_G}, \ \forall g \in \mathcal{G}, \tag{10d}$$

where ω_k is a non-negative weight indicating the priority of user k, and $R_k = \log_2(1+\gamma_k)$ denotes the achievable rate of user k.

To solve this non-convex problem efficiently, we develop an AO framework in the next section, which iteratively updates the beamforming matrix $\mathbf{W}$ and the BD-RIS scattering matrix $\mathbf{\Theta}$ until convergence.

III. Alternating Optimization Framework

Here we adopt an AO framework that iteratively solves two sub-problems: beamforming optimization at the BS and scattering matrix design at the BD-RIS. The beamforming optimization is addressed using the WMMSE method, which effectively handles the SINR coupling and ensures convergence. For scattering matrix optimization, the approach combines fractional programming to reformulate the objective, manifold optimization to handle the unitary constraint, and symmetric unitary projection to enforce the required structure. The following subsections detail the solution procedures for each sub-problem.

A. WMMSE for Beamforming Optimization

When the scattering matrix $\mathbf{\Theta}$ is fixed, the sub-problem with respect to $\mathbf{W}$ is given by

$$\max_{\mathbf{W}} \quad f_{\mathrm{A}_1}(\mathbf{W}) = \sum_{k\in\mathcal{K}} \omega_k R_k \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{W}\|_F^2 \leq P_t. \tag{11b}$$

Based on the WMMSE method proposed in [34], the WSR maximization problem in (11) can be converted to an equivalent weighted mean square error (MSE) minimization problem via the following process. Each user k is equipped with a receiver g_k, which acts as a weight to estimate the transmitted symbol s_k. The estimated symbol $\hat{s}_k$ at user k is expressed as

$$\begin{aligned}\hat{s}_k &= g_k y_k \\ &= g_k \mathbf{c}_k^H \sum_{j\in\mathcal{K}} \mathbf{w}_j s_j + g_k n_k.\end{aligned} \tag{12}$$

The MSE of decoding s_k is calculated as

$$\begin{aligned} e_k &= \mathbb{E}[|\hat{s}_k - s_k|^2], \\ &= |g_k|^2 T_k - 2\Re\left(g_k \mathbf{c}_k^H \mathbf{w}_k\right) + 1, \end{aligned} \tag{13}$$

where $T_k \triangleq \sum_{j\in\mathcal{K}} |\mathbf{c}_k^H \mathbf{w}_j|^2 + \sigma_k^2$ is the average power of the received signal. The solution to the following unconstrained convex optimization problem yields the minimum MSE (MMSE) receiver

$$\begin{aligned} g_k^{\mathrm{MMSE}} &\triangleq \arg\min_{g_k} e_k \\ &= \mathbf{w}_k^H \mathbf{c}_k T_k^{-1}, \end{aligned} \tag{14}$$

where $\partial e_k / \partial g_k = 0$ is the condition of the optimal MMSE receiver. By substituting (14) into (13), the MSE at the output of MMSE receiver is

$$\begin{aligned} e_k^{\mathrm{MMSE}} &\triangleq \mathbb{E}\Big[(g_k^{\mathrm{MMSE}} - s_k)(g_k^{\mathrm{MMSE}} y_k - s_k)^*\Big] \\ &= \left(1 + \frac{|\mathbf{c}_k^H \mathbf{w}_k|^2}{\sum_{j\in\mathcal{K}, j\neq k} |\mathbf{c}_k^H \mathbf{w}_j|^2 + \sigma_k^2}\right)^{-1} \\ &= \left(T_k - |\mathbf{c}_k^H \mathbf{w}_k|^2\right) T_k^{-1}. \end{aligned} \tag{15}$$

Then, the SINR in (9) and achievable rate R_k can be transformed to $\gamma_k = 1/e_k^{\mathrm{MMSE}} - 1$ and $R_k = -\log_2(e_k^{\mathrm{MMSE}})$, respectively. However, the logarithmic rate-MMSE relationships above cannot be used directly for the WSR problem in (11). By introducing an auxiliary variable $\tau_k \in \mathbb{C}$ for the k-th user and defining $\boldsymbol{\tau} = [\tau_1, \ldots, \tau_K]^T$, problem (11) can be transformed into an equivalent form

$$\min_{\mathbf{W}, \boldsymbol{\tau}, \mathbf{g}} \quad \sum_{k\in\mathcal{K}} \omega_k h_k(\mathbf{W}, \boldsymbol{\tau}, \mathbf{g}) \tag{16a}$$

$$\text{s.t.} \quad \|\mathbf{W}\|_F^2 \leq P_t, \tag{16b}$$

where $h_k(\mathbf{W}, \boldsymbol{\tau}, \mathbf{g}) \triangleq \tau_k e_k - \log_2(\tau_k)$, $\forall k \in \mathcal{K}$, is the augmented MMSE and $\mathbf{g} = [g_1, \ldots, g_k]^T$ is the receiver vector. Although the above problem is significantly simpler than the problem (11), it remains challenging to solve. However, the objective function in (16a) is convex with respect to each of the three variables (i.e., $\mathbf{W}$, $\boldsymbol{\tau}$, and $\mathbf{g}$) when the other two are fixed. This property makes the problem more tractable. In the following we present an efficient AO algorithm to solve problem (16) iteratively.

It is worth noting that the auxiliary vector $\boldsymbol{\tau}$ does not appear in the constraints of problem (16). This means that for fixed $\mathbf{W}$ and $\mathbf{g}$, determining the optimal $\boldsymbol{\tau}^{\mathrm{MMSE}}$ becomes an unconstrained optimization problem. Consequently, the optimal solution can be derived by setting the first-order partial derivative of the objective function in (16) with respect to τ_k, $\forall k \in \mathcal{K}$ to zero. This yields the closed-form optimal solution.

$$\tau_k^{\mathrm{MMSE}} = \left(e_k^{\mathrm{MMSE}}\right)^{-1}, \ \forall k \in \mathcal{K}. \tag{17}$$

After determining g_k^{MMSE} and τ_k^{MMSE} for all k, the next step is to optimize the beamforming matrix $\mathbf{W}$. We propose a duality-based method for this optimization. Decomposing problem (16) into sub-problem with respect to $\mathbf{W}$ yields the following

Algorithm 1 Dual-Based WMMSE Algorithm for Calculating $\mathbf{W}$

Input: $\mathbf{c}_k$, $\forall k \in \mathcal{K}$ and μ.
Output: $\mathbf{W}^\star$.

1: Initialize $\mu^{(0)}$, $\mathbf{W}^{(0)}$, and iteration index $t = 0$.
2: Calculate the value of the objective function in (19a) as $\mathrm{WSR}\left(\mathbf{W}^{(0)}\right)$
3: **repeat**
4: set $t = t + 1$;
5: Update $g_k^{(t)}$ by (14).
6: Update $\tau_k^{(t)}$ by (17).
7: Update $\mathbf{W}^{(t)}$ by (23) and search $\mu^{(t)}$ by employing bisection method.
8: **until** $|\mathrm{WSR}\left(\mathbf{W}^{(t)}\right) - \mathrm{WSR}\left(\mathbf{W}^{(t-1)}\right)| \leq \epsilon$.
9: Return $\mathbf{W}^\star$.

sub-problem

$$\mathbf{W}^\star = \arg\min_{\mathbf{W}} \sum_{k\in\mathcal{K}} \omega_k \tau_k e_k, \ \text{s.t. (11b)}. \tag{18}$$

With fixed scattering matrix $\boldsymbol{\Theta}$ and optimal $\boldsymbol{\tau}^{\mathrm{MMSE}}$ and $\mathbf{g}^{\mathrm{MMSE}}$, we focus on the sub-problem expressed in (18) for optimizing beamforming matrix $\mathbf{W}^\star$ at the BS. Since the terms $\omega_k \tau_k |g_k|^2 \sigma_k^2$ and $\omega_k \tau_k$ in the objective function of problem (18) are irrelevant constants with respect to $\mathbf{W}$, they can be omitted, allowing the optimal solution for $\mathbf{W}$ to be determined by solving the following sub-problem.

$$\min_{\mathbf{W}} \quad \sum_{k\in\mathcal{K}} \mathbf{w}_k^H \mathbf{Q} \mathbf{w}_k - \sum_{k\in\mathcal{K}} 2\,\omega_k \tau_k \Re\left(g_k \mathbf{c}_k^H \mathbf{w}_k\right) \tag{19a}$$

$$\text{s.t.} \quad \text{(11b)}. \tag{19b}$$

The matrix $\mathbf{Q}$ is defined as $\mathbf{Q} \triangleq \sum_{k\in\mathcal{K}} \omega_k \tau_k |g_k|^2 \mathbf{c}_k \mathbf{c}_k^H$, where each term $\mathbf{c}_k \mathbf{c}_k^H$ is a rank-one positive semidefinite matrix. Multiplying these matrices by the non-negative scalar coefficients $\omega_k \tau_k |g_k|^2 \geq 0$ preserves their positive semidefinite property, ensuring that $\mathbf{Q}$ is also positive semidefinite. Consequently, the objective function in (19a) is convex. Furthermore, the constraint in (19b) is convex with respect to $\mathbf{W}$. Therefore, problem (19) is a convex optimization problem and can be efficiently solved using convex solvers such as CVX [35]. However, instead of relying on a generic solver, which may incur high computational complexity, we propose a duality-based method to address problem (19) more efficiently. The Lagrangian for problem (19) is expressed as

$$\begin{aligned} \mathcal{L}(\mathbf{W}, \mu) \triangleq &\sum_{k\in\mathcal{K}} \mathbf{w}_k^H \mathbf{Q} \mathbf{w}_k \\ &- \sum_{k\in\mathcal{K}} 2\,\omega_k \tau_k \Re\left(g_k \mathbf{c}_k^H \mathbf{w}_k\right) + \mu\left(\|\mathbf{W}\|_F^2 - P_t\right), \end{aligned} \tag{20}$$

where $\mu \geq 0$ is the Lagrange multiplier associated with the power constraint.

The dual Problem of (19) is defined as

$$\max_{\mu\geq 0} \ g(\mu) = \min_{\mathbf{W}} \ \text{(19a)}, \ \text{s.t.(19b)}. \tag{21}$$

It is clear that any feasible solution to problem (21) is also feasible for problem (19). The relationship between the

optimal values of these two problems is established in the following lemma.

Lemma 1. *Since the original problem (19) is convex and satisfies Slater's condition, strong duality holds. This ensures that the optimal value of the dual problem, i.e.,* $\max_{\mu\geq 0} g(\mu)$*, is equal to the optimal value of the primal problem (19).*

Proof. This lemma directly follows from *Proposition* 4 in [36] and is omitted here for brevity. □

The Karush-Kuhn-Tucker (KKT) conditions must hold at the optimal solutions, hence according to stationarity and complementary slackness conditions, we have the following

$$\nabla_{\mathbf{w}_k}\mathcal{L}\left(\mathbf{W},\mu\right)=0, \tag{22a}$$
$$\mu\left(\|\mathbf{W}\|_F^2-P_t\right)=0. \tag{22b}$$

From the stationarity condition in (22a), with a fixed μ, the optimal solution of $\mathbf{w}_k, \forall k\in\mathcal{K}$ is calculated by

$$\mathbf{w}_k\left(\mu\right)=\omega_k\tau_k g_k^*\Big(\mathbf{Q}+\mu\mathbf{I}_M\Big)^{-1}\mathbf{c}_k,\ \forall k\in\mathcal{K}. \tag{23}$$

Then, the dual variable μ should be determined for satisfying the complementary slackness condition in (22b). We search μ for two cases, i.e., $\mu=0$ and $\mu>0$. When $\mu=0$, the constraint $\|\mathbf{W}\|_F^2\leq P_t$ is inactive (passive), meaning that the unconstrained solution satisfies the power budget. In this case, the optimal solution simplifies to

$$\mathbf{w}_k=\mathbf{w}_k\left(0\right)=\omega_k\tau_k g_k^*\mathbf{Q}^{-1}\mathbf{c}_k. \tag{24}$$

When $\mu\geq 0$, the power constraint is active, and the optimal solution $\mathbf{w}_k$ is adjusted to satisfy

$$\|\mathbf{W}\left(\mu^\star\right)\|_F^2=P_t, \tag{25}$$

where $\mu^\star$ is the optimal Lagrange multiplier obtained from simple bisection search method.

B. Scattering Matrix Optimization

For the fixed $\mathbf{W}$, the problem (10) is simplified to

$$\max_{\mathbf{\Theta}}\quad f_{\mathrm{A}_2}(\mathbf{\Theta})=\sum_{k\in\mathcal{K}}\omega_k\log_2\left(1+\gamma_k\right) \tag{26a}$$
$$\text{s.t.}\quad \mathbf{\Theta}=\mathsf{diag}\left(\mathbf{\Theta}_1,\ldots,\mathbf{\Theta}_G\right), \tag{26b}$$
$$\mathbf{\Theta}_g=\mathbf{\Theta}_g^T,\ \mathbf{\Theta}_g^H\mathbf{\Theta}_g=\mathbf{I}_{N_G},\ \forall g\in\mathcal{G}. \tag{26c}$$

The logarithmic fractional form of WSR and the quadratic equality constraint lead to the non-convexity of problem (26). Nevertheless, we show that it can be tackled based on Lagrangian Dual Transform [37], Quadratic Transform [38], manifold algorithm [17], [39], and symmetric unitary projection [40]. Applying the Lagrangian Dual Transform proposed in [37], $f_{\mathrm{A}_2}(\mathbf{\Theta})$ can be equivalently transformed into the following form

$$f_{\boldsymbol{\alpha}}\left(\mathbf{\Theta},\boldsymbol{\alpha}\right)=\sum_{k\in\mathcal{K}}\Big(\omega_k\log_2(1+\alpha_k)-\omega_k\alpha_k+\frac{\omega_k(1+\alpha_k)\gamma_k}{1+\gamma_k}\Big), \tag{27}$$

where $\boldsymbol{\alpha}\triangleq[\alpha_1,\ldots,\alpha_K]^T\in\mathbb{R}^K$ is an auxiliary vector. Next, we apply Quadratic Transform [38] on the fractional term in (27) and reformulate the objective function $f_{\boldsymbol{\alpha}}\left(\mathbf{\Theta},\boldsymbol{\alpha}\right)$ as:

$$\begin{aligned}f_{\boldsymbol{\beta}}\left(\mathbf{\Theta},\boldsymbol{\alpha},\boldsymbol{\beta}\right)=\sum_{k\in\mathcal{K}}\Big(&\omega_k\log_2\left(1+\alpha_k\right)-\omega_k\alpha_k\\&+2\sqrt{\omega_k\left(1+\alpha_k\right)}\Re\left(\beta_k^*\mathbf{c}_k^H\mathbf{w}_k\right)\\&-|\beta_k|^2\sum_{j\in\mathcal{K}}\left(|\mathbf{c}_k^H\mathbf{w}_j|^2+\sigma_k^2\right)\Big),\end{aligned} \tag{28}$$

where $\boldsymbol{\beta}\triangleq[\beta_1,\ldots,\beta_K]^T\in\mathbb{C}^K$ denotes another auxiliary vector. Now, with auxiliary vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, the sub-problem (26) can be transformed into:

$$\max_{\mathbf{\Theta},\boldsymbol{\alpha},\boldsymbol{\beta}}\quad f_{\boldsymbol{\beta}}\left(\mathbf{\Theta},\boldsymbol{\alpha},\boldsymbol{\beta}\right) \tag{29a}$$
$$\text{s.t.}\quad \text{(26b), (26c).} \tag{29b}$$

For fixed $\mathbf{\Theta}$, (29) is an unconstrained convex optimization problem with respect to $\boldsymbol{\alpha}$ (or $\boldsymbol{\beta}$) whose solution can be derived by setting $\frac{\partial f_{\boldsymbol{\beta}}(\mathbf{\Theta},\boldsymbol{\alpha},\boldsymbol{\beta})}{\partial\boldsymbol{\alpha}}=\mathbf{0}$ (or $\frac{\partial f_{\boldsymbol{\beta}}(\mathbf{\Theta},\boldsymbol{\alpha},\boldsymbol{\beta})}{\partial\boldsymbol{\beta}}=\mathbf{0}$). Then the optimal solutions for auxiliary variables α_k and β_k can be obtained as:

$$\alpha_k^\star=\gamma_k,\ \forall k\in\mathcal{K}, \tag{30}$$

$$\beta_k^\star=\frac{\sqrt{\omega_k\left(1+\alpha_k\right)}\mathbf{c}_k^H\mathbf{w}_k}{\sum_{j\in\mathcal{K}}|\mathbf{c}_k^H\mathbf{w}_j|^2+\sigma_k^2},\ \forall k\in\mathcal{K}. \tag{31}$$

When $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are determined, $f_{\boldsymbol{\beta}}\left(\mathbf{\Theta},\boldsymbol{\alpha},\boldsymbol{\beta}\right)$ can be reformulated to a new function $f_1\left(\mathbf{\Theta}\right)=\sum_{k\in\mathcal{K}}\Big(2\Re\big(\tilde{\beta}_k^*\mathbf{h}_k^H\mathbf{\Theta}\tilde{\mathbf{h}}_k\big)-|\beta_k|^2\sum_{j\in\mathcal{K}}|\mathbf{h}_k^H\mathbf{\Theta}\tilde{\mathbf{h}}_j|^2\Big)$, where $\tilde{\beta}_k\triangleq\sqrt{\omega_k\left(1+\alpha_k\right)}\beta_k$, $\tilde{\mathbf{h}}_k\triangleq\mathbf{H}\mathbf{w}_k,\ \forall k\in\mathcal{K}$. Therefore, the sub-problem with respect to $\mathbf{\Theta}$ can be written as [17]

$$\min_{\mathbf{\Theta}}\quad f_2\left(\mathbf{\Theta}\right)=\mathsf{Tr}\left(\mathbf{\Theta}\mathbf{Y}\mathbf{\Theta}^H\mathbf{Z}\right)-2\Re\left(\mathsf{Tr}\left(\mathbf{\Theta}\mathbf{X}\right)\right) \tag{32a}$$
$$\text{s.t.}\quad \mathbf{\Theta}=\mathsf{diag}\left(\mathbf{\Theta}_1,\ldots,\mathbf{\Theta}_G\right), \tag{32b}$$
$$\mathbf{\Theta}_g=\mathbf{\Theta}_g^T,\ \mathbf{\Theta}_g^H\mathbf{\Theta}_g=\mathbf{I}_{N_G},\ \forall g\in\mathcal{G}, \tag{32c}$$

where

$$\mathbf{X}\triangleq\sum_{k\in\mathcal{K}}\Big(\tilde{\beta}_k^*\tilde{\mathbf{h}}_k\mathbf{h}_k^H-|\beta_k|^2\sum_{j\in\mathcal{K}}\left(\mathbf{d}_k^H\mathbf{w}_j\right)^*\tilde{\mathbf{h}}_j\mathbf{h}_k^H\Big),$$
$$\mathbf{Y}\triangleq\sum_{j\in\mathcal{K}}\tilde{\mathbf{h}}_j\tilde{\mathbf{h}}_j^H,$$
$$\mathbf{Z}\triangleq\sum_{k\in\mathcal{K}}|\beta_k|^2\mathbf{h}_k\mathbf{h}_k^H.$$

To handle the symmetric unitary constraint (32c), we first relax it to require only being unitary, i.e., $\mathbf{\Theta}_g^H\mathbf{\Theta}_g=\mathbf{I}_{N_G},\forall g\in\mathcal{G}$. By employing manifold optimization techniques, we reformulate (32) as an unconstrained problem on a manifold [17], [41]. For notational brevity we consider the fully connected case ($G=1$); then the proposed solution can be extended to group connected ($G=N/N_G$) and single connected ($G=N$) architectures. In this special case, $\mathbf{\Theta}$ lies on the N^2-dimensional complex Stiefel manifold $\mathcal{M}=\left\{\mathbf{\Theta}\in\mathbb{C}^{N\times N}\mid\mathbf{\Theta}^H\mathbf{\Theta}=\mathbf{I}_N\right\}$, therefore the problem (32) reduces to [17]

$$\mathbf{\Theta}^\star=\arg\min_{\mathbf{\Theta}\in\mathcal{M}}f_2\left(\mathbf{\Theta}\right), \tag{33}$$

Algorithm 2 Alternating Optimization to solve (10)

Initialize $i \leftarrow 0, \mathbf{\Theta}_0, \mathbf{W}_0, \eta_0$.
repeat
  Update $\mathbf{W}^{(i+1)}$ using Algorithm 1.
  Optimize $\mathbf{\Theta}$ via manifold-based optimization:
  while no convergence of $\|\nabla_{\mathcal{M}} f_2(\mathbf{\Theta}_i)\|_F$
    $i \leftarrow i+1$;
    Calculate δ_i using backtracking algorithms [39];
    Update $\mathbf{\Theta}_i$ using (38).
    Update κ_i using (37).
    Update ξ_i using (36).
  end while
  Obtain $\mathbf{\Theta}^\star$ by enforcing BD-RIS constraints:
  if $G = 1$ (Fully-Connected)
    $\mathbf{\Theta}^\star_{\text{FC}} = \mathsf{symuni}(\mathbf{\Theta}_{i+1})$.
  elseif $G = N$ (Single-Connected)
    $\mathbf{\Theta}^\star_{\text{SC}} = \mathsf{diag}(e^{\mathrm{1j}\angle x_1}, \ldots, e^{\mathrm{1j}\angle x_N})$.
  else (Group-Connected)
    Block-diagonalize $\mathbf{\Theta}_{i+1}$ using (40).
    $\mathbf{\Theta}^\star_{\text{GC}} = \mathsf{diag}(\mathsf{symuni}(\mathbf{X}_1), \ldots, \mathsf{symuni}(\mathbf{X}_G))$.
until $|\text{WSR}(t) - \text{WSR}(t-1)| \le \epsilon$.
Return $\mathbf{W}^\star, \mathbf{\Theta}^\star$.

which can be tackled using *Riemannian* optimization methods [17], [41].

Essentially, we adapt Euclidean conjugate-gradient (CG) techniques to manifold $\mathcal{M}$ by applying the necessary projections. To this end, we first calculate Euclidean gradient of $f_2(\mathbf{\Theta})$ which is obtained as

$$\nabla f_2(\mathbf{\Theta}) = 2\mathbf{Z}\mathbf{\Theta}\mathbf{Y} - 2\mathbf{X}^H. \tag{34}$$

Next, we define the tangent space at point $\mathbf{\Theta}$ on $\mathcal{M}$ as $\mathrm{T}_{\mathbf{\Theta}}\mathcal{M} = \left\{\mathbf{T} \in \mathbb{C}^{N\times N} \mid \Re\left(\mathbf{\Theta}^H\mathbf{T} = \mathbf{0}_N\right)\right\}$ which comprises of all the tangent vectors indicating all possible directions this point can move to. By projecting the Euclidean gradient in (34) onto $\mathrm{T}_{\mathbf{\Theta}}\mathcal{M}$, we obtain the Riemannian gradient, which points towards the steepest descent direction of $f_2(\mathbf{\Theta})$, given by [17], [39], [41]

$$\begin{aligned}\nabla_{\mathcal{M}} f_2(\mathbf{\Theta}) &= \mathsf{Pr}_{\mathbf{\Theta}}\left(\nabla f_2(\mathbf{\Theta})\right) \\ &= \nabla f_2(\mathbf{\Theta}) - \mathbf{\Theta}\,\mathsf{chdiag}\left(\mathbf{\Theta}^H\nabla f_2(\mathbf{\Theta})\right),\end{aligned} \tag{35}$$

where $\mathsf{Pr}_{\mathbf{\Theta}}(\cdot)$ denotes the projection function, and $\mathsf{chdiag}(\cdot)$ extracts and rearranges the diagonal elements into a diagonal matrix. With the Riemannian gradient in hand, we apply the CG algorithm on the manifold. At the i-th iteration of the CG method, we successively do the following steps

Step 1: Find the descent direction in $\mathrm{T}_{\mathbf{\Theta}}\mathcal{M}$ as

$$\eta_i = -\nabla_{\mathcal{M}} f_2(\mathbf{\Theta}_i) + \kappa_i \mathsf{Pr}_{\mathbf{\Theta}_i}(\eta_{i-1}), \tag{36}$$

where κ_i is the CG update parameter based on the Riemannian Polak-Ribiére formula as [39]

$$\kappa_i = \frac{\Re\left(\mathsf{Tr}\left(\mathbf{F}_i^H\left(\mathbf{F}_i - \mathsf{Pr}_{\mathbf{\Theta}_i}(\mathbf{F}_{i-1})\right)\right)\right)}{\mathsf{Tr}\left(\mathbf{F}_{i-1}^H\mathbf{F}_{i-1}\right)}, \tag{37}$$

with $\mathbf{F}_i \triangleq \nabla_{\mathcal{M}} f_2(\mathbf{\Theta}_i)$.

Step 2: Update $\mathbf{\Theta}$ by performing a retraction to remain on manifold $\mathcal{M}$ as [17], [28]

$$\mathbf{\Theta}_{i+1} = (\mathbf{\Theta}_i + \delta_i\eta_i)\left(\mathbf{I}_N + \delta_i^2\eta_i^H\eta_i\right)^{-1/2}, \tag{38}$$

where δ_i is the step size determined by a backtracking algorithm [39]. A local optimal solution of problem (33), i.e., $\mathbf{\Theta}^\star$, can be obtained by appropriately initializing $\mathbf{\Theta}$ and then iteratively updating η_i, κ_i, $\mathbf{\Theta}_i$ and δ_i until convergence.[2] Note that the optimal solution $\mathbf{\Theta}^\star$ of problem (26) may not satisfy both the symmetric and unitary constraints of the BD-RIS. To ensure feasibility, we employ the symmetric unitary projection method proposed in [40], which enforces these constraints for fully-, group-, and single-connected RIS architectures. Specifically, we first symmetrize $\mathbf{\Theta}^\star$ as $\tilde{\mathbf{\Theta}} = \frac{\mathbf{\Theta}^\star + (\mathbf{\Theta}^\star)^T}{2}$ and assume its rank is R. The singular value decomposition (SVD) of $\tilde{\mathbf{\Theta}}$ is $\tilde{\mathbf{\Theta}} = \mathbf{U}\mathbf{S}\mathbf{V}^H$, where $\mathbf{U} = [\mathbf{U}_R, \mathbf{U}_{N-R}]$ and $\mathbf{V} = [\mathbf{V}_R, \mathbf{V}_{N-R}]$. For the fully-connected case, a feasible scattering matrix $\mathbf{\Theta}^\star_{\text{FC}}$ is computed as [40]

$$\mathbf{\Theta}^\star_{\text{FC}} = \mathsf{symuni}(\mathbf{\Theta}^\star) \triangleq \widehat{\mathbf{U}}\mathbf{V}^H, \tag{39}$$

where $\widehat{\mathbf{U}} = [\mathbf{U}_R, \mathbf{V}^*_{N-R}]$. For the group-connected case, we first block-diagonalize $\mathbf{\Theta}^\star$ and then apply symmetric unitary projection to each block. The block diagonalization is performed as

$$\mathsf{blkdiag}(\mathbf{\Theta}^\star) = \mathsf{diag}(\mathbf{1}, \ldots, \mathbf{1}) \odot \mathbf{\Theta}^\star, \tag{40}$$

where $\mathbf{1}$ is a N_g-dimensional square matrix filled with ones. Defining $\mathsf{diag}(\mathbf{X}_1, \ldots, \mathbf{X}_G) = \mathsf{blkdiag}(\mathbf{\Theta}^\star)$, a feasible scattering matrix for group-connected case is given by

$$\mathbf{\Theta}^\star_{\text{GC}} = \mathsf{diag}\left(\mathsf{symuni}(\mathbf{X}_1), \ldots, \mathsf{symuni}(\mathbf{X}_G)\right). \tag{41}$$

The single-connected RIS can be seen as a special case of the group-connected RIS, where each group consists of a single element, i.e., $G = N$. By setting $N_g = 1, \forall g \in \mathcal{G}$, (41) simplifies to a feasible scattering matrix for the single-connected case, given by

$$\mathbf{\Theta}^\star_{\text{SC}} = \mathsf{diag}\left(e^{\mathrm{1j}\angle x_1}, \ldots, e^{\mathrm{1j}\angle x_N}\right), \tag{42}$$

where each block $\mathbf{X}_n$ reduces to scalar x_n for $n \in \{1, \ldots, N\}$. The proposed AO algorithm to jointly optimize the beamforming matrix and the scattering matrix is summarized in Algorithm 2.

C. Complexity Analysis

In this subsection, we briefly analyze the total computational complexity of the Algorithm 2. The outer loop consists of two sub-problems for optimizing $\mathbf{W}$ and $\mathbf{\Theta}$, each requiring an iterative update procedure. Specifically, the WMMSE algorithm involves matrix inversion operations in all the three update steps (Equations (14), (17), and (23)), leading to a total complexity of $\mathcal{O}\{KM^3\}$. Additionally, the bisection search in (23) requires I_μ iterations. Consequently, the overall complexity of the WMMSE algorithm is $\mathcal{O}\{I_1KM^3\}$, where I_1 is the number of iterations across the three update steps.

[2]It is worth noting that this work considers both symmetric and unitary BD-RIS constraints, whereas [17], [28] assumes only unitary and implicitly symmetric designs.

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Number of antennas at the BS	$M = 32$
BS and RIS heights	$h_{\mathrm{BS}} = 6\,\mathrm{m}$, $h_{\mathrm{RIS}} = 3\,\mathrm{m}$
User heights	$h_{\mathrm{u}} = 1.5\,\mathrm{m}$
LoS path loss parameters	$\mathrm{PL}_0 = 61.4$, $b = 2.0$, $\sigma_\zeta = 5.8$
NLoS path loss parameters	$\mathrm{PL}_0 = 72.0$, $b = 2.92$, $\sigma_\zeta = 8.7$
Transmit power at the BS	$P_t = 30$ dBm
Noise power	$\sigma^2 = -85$ dBm
Convergence threshold	$\epsilon = 10^{-3}$

For the scattering matrix optimization, the manifold version of the CG method (steps 5–9 in Algorithm 2) incurs a complexity of $\mathcal{O}\{I_2 N^3\}$ for fully-connected BD-RIS, where I_2 denotes the number of CG iterations. This complexity reduces to $\mathcal{O}\{I_2 G N_g^2\}$ for group-connected BD-RIS and further to $\mathcal{O}\{I_2 G\}$ for single-connected BD-RIS. The symmetric unitary projection step primarily depends on the SVD of the symmetric matrix $\tilde{\boldsymbol{\Theta}}$. Its complexity is $\mathcal{O}\{N^3\}$ for fully-connected BD-RIS, $\mathcal{O}\{N N_g^2\}$ for group-connected BD-RIS and $\mathcal{O}\{N\}$ for single-connected BD-RIS. Thus, the total complexity of Algorithm 2 is $\mathcal{O}\{I_0\left(I_\mu I_1 K M^3 + I_2 N^3\right)\}$ for fully-connected BD-RIS, $\mathcal{O}\{I_0\left(I_\mu I_1 K M^3 + I_2 G N_g^3\right)\}$ for group-connected BD-RIS, and $\mathcal{O}\{I_0\left(I_\mu I_1 K M^3 + I_2 G\right)\}$ for single-connected BD-RIS. Here, I_0 denotes the total number of outer-loop iterations.

IV. Numerical Results

In this section, we present numerical results that evaluate the performance of the proposed AO algorithm across different BD-RIS architectures. We consider a three-dimensional setup, as illustrated in Fig. 2, in which the BS and RIS are located at $[0, 0]$ m and $[60, 0]$ m, respectively. A total of K users are uniformly distributed within a circular area centered at $[70, 0]$ m with a radius of $r = 10$ m. All links follow the geometric SV channel model, as described in Section II-A. The number of NLoS components is set to 4 for each channel, resulting in $L_{\mathrm{BU}} = 4$, $L_{\mathrm{RU}} = 5$, and $L_{\mathrm{BR}} = 5$. The Rician factor, defined as the ratio of the energy in the LoS path to the total energy in the NLoS paths, is set to 13.2 dB [11]. Azimuth and elevation AoA and AoD are assumed to be uniformly distributed over $[-\pi, \pi)$ and $[-\pi/2, \pi/2)$, respectively, except for the LoS AoA, which is determined by the user's actual location [31].

The path loss PL is modeled as

$$\mathrm{PL} = \mathrm{PL}_0 + 10b\log_{10}(D) + \zeta \text{ dB},$$

where D is the link distance and $\zeta \sim \mathcal{N}(0, \sigma_\zeta^2)$ denotes log-normal shadow fading. Following the measurements in [7, Table I], different values are used for LoS and NLoS links as listed in Table II. A wall penetration loss of 6.8 dB is added to account for indoor entry through drywall [5]. The rest of the simulation parameters are summarized in Table II.

We evaluate the WSR performance of our proposed Algorithm 2 across different BD-RIS architectures, i.e., fully-, group-, and single-connected BD-RIS, where the group size is set to $N_G = 4$. We also compare its performance and runtime against three baseline methods:

- Two-stage FP algorithm [40]: Optimizes the scattering matrix in a heuristic way and uses fractional programming for beamforming.
- WMMSE-CVX MO [42]: Employs manifold optimization

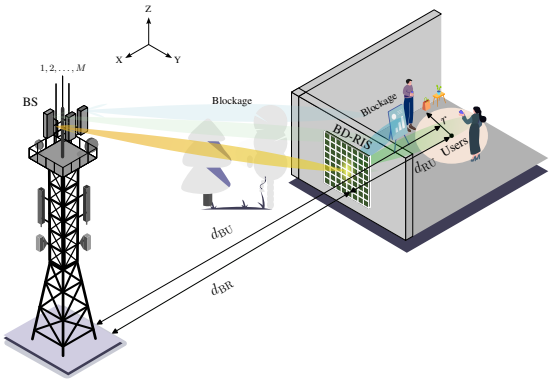


Fig. 2. System model of BD-RIS-assisted outdoor-to-indoor mmWave communication with different BD-RIS architectures: (a) fully-connected, (b) group-connected, (c) single-connected.

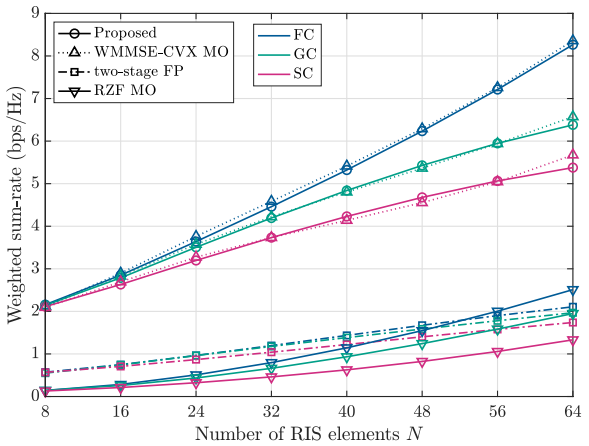


Fig. 3. WSR versus number of RIS elements for fully-connected "FC", group-connected "GC", and single-connected "SC" BD-RIS ($K = 2$, $N_G = 4$, $\eta = 0$).

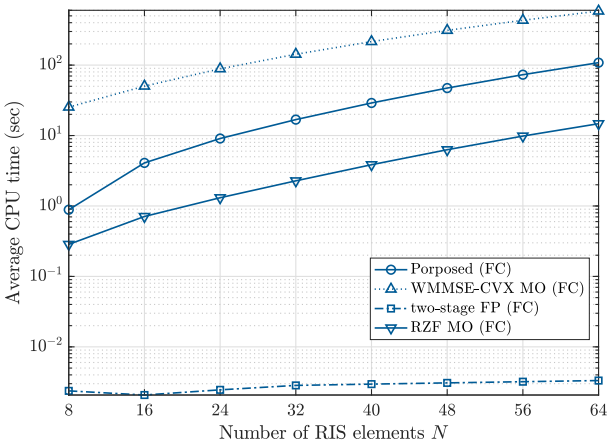
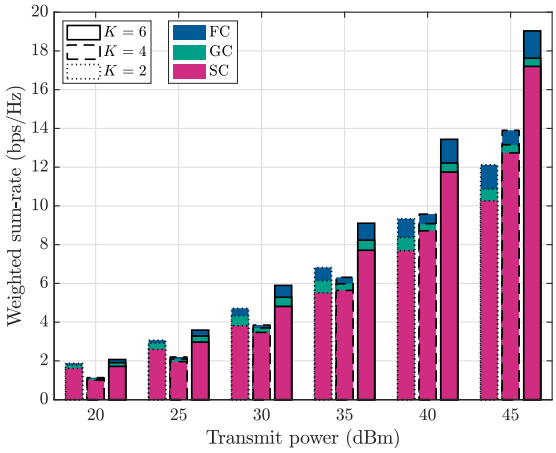
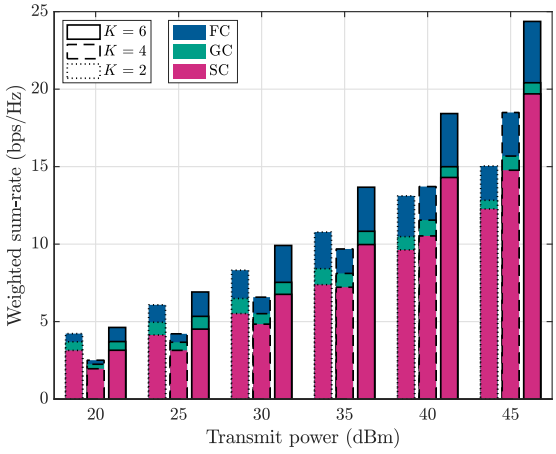


Fig. 4. Average CPU time versus number of RIS elements for fully-connected "FC" BD-RIS ($K = 2$, $\eta = 0$).



(a) $N = 32$



(b) $N = 64$

Fig. 5. WSR versus transmit power for fully-connected "FC", group-connected "GC", and single-connected "SC" BD-RIS, evaluated for $K = 6$, $K = 4$, and $K = 2$ users ($N_G = 4$, $\eta = 0$).

for the scattering matrix and WMMSE-CVX for beamforming.

- RZF MO [40], [43]: Uses manifold optimization for the scattering matrix and regularized zero-forcing (RZF) for beamforming.

Figs. 3 and 4 illustrates the WSR performance and computational efficiency of Algorithm 2 as a function of the number of RIS elements N. When the number of RIS elements is small ($N \leq 16$), the performance gap among BD-RIS architectures remains negligible. However, as N increases ($N \geq 32$), the performance gap widens. Specifically, at $N = 64$, the WSR achieved by fully-connected BD-RIS is 21% and 43% higher than that of group-connected BD-RIS and single-connected BD-RIS, respectively. Our proposed algorithm achieves a WSR comparable to WMMSE-CVX MO but with significantly lower computational complexity. Both approaches outperform the RZF MO algorithm, indicating that RZF is less effective in exploiting the rank-one structure of mmWave channels. Moreover, the two-stage FP algorithm fails to capture the performance differences among BD-RIS architectures in rank-one mmWave channels. This is because it heuristically determines the scattering matrix $\boldsymbol{\Theta} \in \mathbb{C}^{N \times N}$

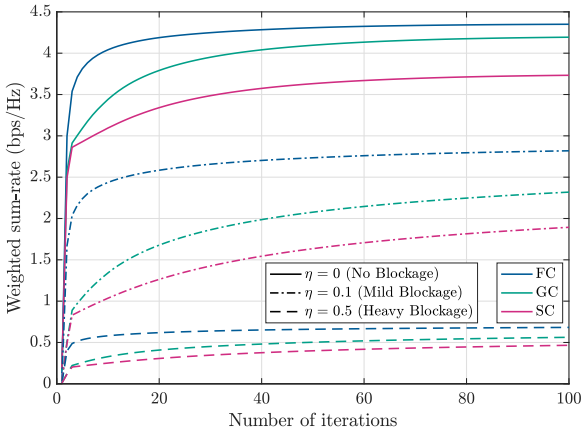


Fig. 6. Convergence behavior of the proposed algorithm for fully-connected "FC", group-connected "GC", and single-connected "SC" BD-RIS. ($N = 32$, $K = 2$, $N_G = 4$).

and keeps it fixed during WSR maximization. As a result, the algorithm does not adapt to the rank of the channel, thereby limiting the performance benefits of fully-connected BD-RIS compared to group-connected and single-connected architectures.

As the transmit power increases from 20 dBm to 45 dBm, all three multiuser configurations with $K = 2$, $K = 4$, and $K = 6$ exhibit a consistent upward trend in WSR, as shown in Fig. 5. In the low-transmit-power regime (below approximately 40 dBm), the grouped bars exhibit comparable performance because the received noise term in the MMSE receiver update (15) dominates, and the weight update (17) assigns smaller weights to streams with larger mean-square error, which cause the precoder (23) to spread power conservatively across users. In the high-transmit-power regime (beyond approximately 40 dBm), the system becomes interference-limited. In this regime, the weights of low-MSE users increases, the precoder allocates more along the corresponding high channel-gain directions, and the WSR is highest for $K = 6$, followed by $K = 4$, and then $K = 2$. This trend agrees with the findings in [44], which show that serving more users becomes advantageous at high SNR, where interference is the primary bottleneck. The impact of enhanced passive beamforming becomes evident as the number of RIS elements N increases, as seen by comparing performance across different values of N. For example, for $K = 6$, increasing the number of RIS elements from $N = 32$ (see Fig. 5(a)) to $N = 64$ (see Fig. 5(b)) raises the WSR gain of fully-connected BD-RIS over single-connected from 20.66% to 46.74% at 20 dBm, and from 10.65% to 23.73% at 45 dBm. This consistent improvement with larger RIS highlights the system's ability to enhance multiuser transmission by leveraging additional degrees of freedom, as also noted in [28].

The convergence behavior of the proposed AO algorithm for the fully-, group-, and single-connected BD-RIS architectures is illustrated in Fig. 6. Even as the blockage density increases from no blockage ($\eta = 0$) to mild blockage ($\eta = 0.1$) and heavy blockage ($\eta = 0.5$), the algorithm consistently converges to a stable WSR within a limited number of iter-

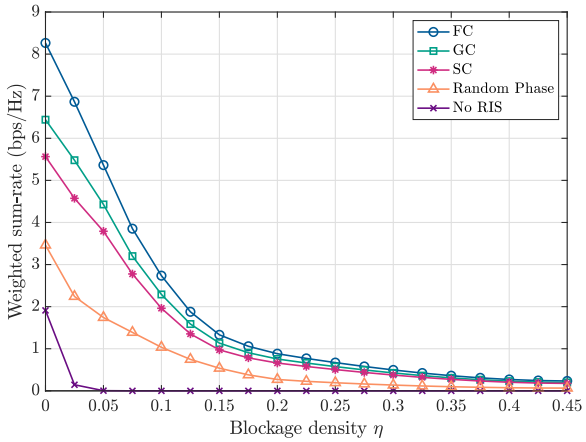


Fig. 7. WSR as a function of increasing blockage density η for fully-connected "FC", group-connected "GC", and single-connected "SC" BD-RIS architectures, along with benchmark schemes using random phase configuration and no RIS. ($N = 64$, $K = 2$, and $N_G = 4$).

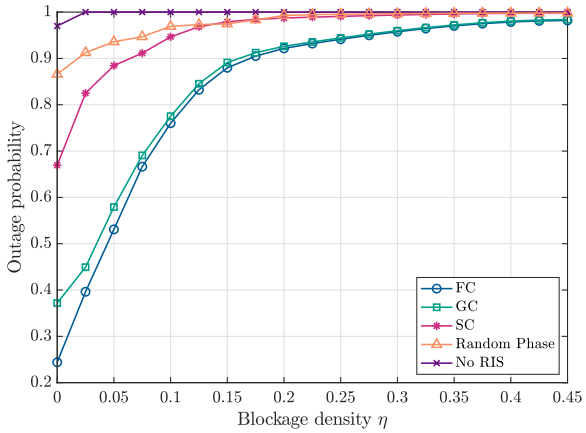


Fig. 8. Outage probability as a function of blockage density η for fully-connected "FC", group-connected "GC", and single-connected "SC" BD-RIS architectures, together with benchmark schemes using random phase configuration and no RIS ($N = 64$, $K = 2$, and $N_G = 4$).

ations, demonstrating its robustness against varying blockage conditions.

The WSR performance under varying blockage density η (see (5)) is evaluated in Fig. 7. As η increases from 0 to 0.45, all BD-RIS-assisted schemes experience a decline in performance, though at different rates. The fully-connected architecture exhibits the most resilience to blockages, while the group-connected and single-connected architectures show progressively steeper declines. The no-RIS baseline quickly collapses, underscoring the importance of BD-RIS for maintaining high data rates in densely obstructed environments.

We define the indicator function $\mathbb{1}\{\cdot\}$ as

$$\mathbb{1}\{R_i(\eta) < \tau\} = \begin{cases} 1 & \text{if } R_i(\eta) < \tau, \\ 0 & \text{otherwise}, \end{cases} \tag{43}$$

where $R_i(\eta)$ denotes the WSR in the i-th trial at blockage density η, and τ is a predefined rate threshold set to $\tau = 5.7$ according to [11], [45]. Based on this definition, the empirical

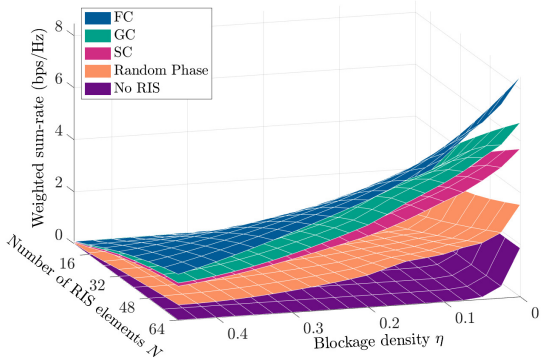


Fig. 9. Comparison of WSR across fully-connected "FC", group-connected "GC", and single-connected "SC" BD-RIS architectures, relative to benchmark schemes with random phase and no RIS, under varying blockage density η and number of RIS elements N ($N = 64$, $K = 2$, and $N_G = 4$).

outage probability at density η is computed as

$$\begin{aligned} \hat{P}_{\text{out}}(\eta;\tau) &= \Pr\{R(\eta) < \tau\} \\ &\approx \frac{1}{T}\sum_{i=1}^{T} \mathbb{1}\{R_i(\eta) < \tau\}, \end{aligned} \tag{44}$$

where $T = 1000$ denotes the total number of Monte Carlo trials.

The resulting outage probability curves are presented in Fig. 8. As the blockage density increases, all curves exhibit a gradual rise from near zero to near one, while the system without RIS reaches outage almost immediately. The fully-connected RIS consistently achieves the lowest outage across all values of η, followed by the group-connected and single-connected architectures.

To further support this trend, the WSR values reported in Fig. 9 demonstrate the performance advantage of BD-RIS architectures under both mild and more severe blockage. Under mild blockage, i.e., $\eta = 0.1$, increasing the number of RIS elements from $N = 32$ to $N = 64$ nearly doubles the WSR for the fully-connected architecture, yielding a 99.52% improvement, while the group-connected and single-connected architectures achieve 66.74% and 66.1% gains, respectively. Under more severe blockage, i.e., $\eta = 0.4$, the gains are still substantial: 91.43% for fully-connected, 75.95% for group-connected, and 68.1% for single-connected. At $N = 64$ and $\eta = 0.1$, the fully-connected RIS provides 25.05% and 48.06% higher WSR than the group-connected and single-connected architectures, respectively, and outperforms the random-phase scheme by 211.27%. The group-connected and single-connected architectures also offer significant gains over the random-phase scheme, with 148.92% and 110.22% improvements, respectively. These results collectively demonstrate the effectiveness of BD-RIS in sustaining high data rates even as blockage conditions become more pronounced.

V. Conclusion and Future Work

This paper investigated a BD-RIS-assisted outdoor-to-indoor mmWave MU-MISO system, where the goal is to

maximize the WSR by optimizing the BS beamforming matrix and the BD-RIS scattering matrix, subject to transmit power and symmetric unitary constraints of the BD-RIS. To tackle this non-convex problem, we proposed an efficient AO algorithm applicable to fully-, group-, and single-connected BD-RIS architectures. Simulation results confirm that the proposed AO algorithm achieves near-optimal WSR with reduced complexity and reveals the performance trade-offs among different BD-RIS architectures. Notably, the fully-connected BD-RIS consistently delivers the highest WSR across various blockage scenarios, while all BD-RIS designs maintain strong robustness and deliver substantial WSR gains over baseline schemes. These results demonstrate the practical potential of BD-RIS in enhancing link reliability for outdoor-to-indoor mmWave communications.

As the number of indoor users scales up, potentially reaching 50 to 100 users instead of a few, can a single BD-RIS still provide sufficient signal support to meet their data rate requirements? If not, how many BD-RIS units are needed, and where should they be placed to maximize coverage and system efficiency? These questions motivate our future work, which aims to extend the current single-BD-RIS setup to scenarios involving multiple BD-RIS units deployed across building exteriors. This extension introduces new challenges, such as jointly optimizing the number, spatial placement, and scattering matrices of multiple BD-RIS units under fixed budgets on the number of RIS elements and the BS transmit power. Additionally, we will investigate how user density and blockage conditions influence deployment strategies and system performance. This study is expected to offer valuable guidelines for practical and scalable BD-RIS deployment in real-world outdoor-to-indoor mmWave networks.

REFERENCES

[1] D. Shakya, D. Chizhik, J. Du, R. A. Valenzuela, and T. S. Rappaport, "Dense urban outdoor-indoor coverage from 3.5 to 28 ghz," in *ICC 2022 - IEEE International Conference on Communications*, 2022, pp. 932–937.

[2] D. W. K. Ng, M. Breiling, C. Rohde, F. Burkhardt, and R. Schober, "Energy-efficient 5g outdoor-to-indoor communication: Sudas over licensed and unlicensed spectrum," *IEEE Transactions on Wireless Communications*, vol. 15, no. 5, pp. 3170–3186, 2016.

[3] C. Umit Bas *et al.*, "Outdoor to indoor propagation channel measurements at 28 ghz," *IEEE Transactions on Wireless Communications*, vol. 18, no. 3, pp. 1477–1489, 2019.

[4] M. K. Samimi, G. R. MacCartney, S. Sun, and T. S. Rappaport, "28 ghz millimeter-wave ultrawideband small-scale fading models in wireless channels," in *2016 IEEE 83rd Vehicular Technology Conference (VTC Spring)*, 2016, pp. 1–6.

[5] H. Zhao, R. Mayzus, S. Sun, M. Samimi, J. K. Schulz, Y. Azar, K. Wang, G. N. Wong, F. Gutierrez, and T. S. Rappaport, "28 ghz millimeter wave cellular communication measurements for reflection and penetration loss in and around buildings in new york city," in *2013 IEEE International Conference on Communications (ICC)*, 2013, pp. 5163–5167.

[6] M. Nemati, B. Maham, S. R. Pokhrel, and J. Choi, "Modeling ris empowered outdoor-to-indoor communication in mmwave cellular networks," *IEEE Transactions on Communications*, vol. 69, no. 11, pp. 7837–7850, 2021.

[7] M. R. Akdeniz *et al.*, "Millimeter wave channel modeling and cellular capacity evaluation," *IEEE Journal on Selected Areas in Communications*, vol. 32, no. 6, pp. 1164–1179, 2014.

[8] A. Alkhateeb, J. Mo, N. Gonzalez-Prelcic, and R. W. Heath, "Mimo precoding and combining solutions for millimeter-wave systems," *IEEE Communications Magazine*, vol. 52, no. 12, pp. 122–131, 2014.

[9] H. Guo, Y.-C. Liang, J. Chen, and E. G. Larsson, "Weighted sum-rate maximization for reconfigurable intelligent surface aided wireless networks," *IEEE Transactions on Wireless Communications*, vol. 19, no. 5, pp. 3064–3076, 2020.

[10] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Transactions on Wireless Communications*, vol. 18, no. 11, pp. 5394–5409, 2019.

[11] P. Wang, J. Fang, X. Yuan, Z. Chen, and H. Li, "Intelligent reflecting surface-assisted millimeter wave communications: Joint active and passive precoding design," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 12, pp. 14 960–14 973, 2020.

[12] H. Yang, S. Kim, H. Kim, S. Bang, Y. Kim, S. Kim, K. Park, D. Kwon, and J. Oh, "Beyond limitations of 5g with ris: Field trial in a commercial network, recent advances, and future directions," *IEEE Communications Magazine*, vol. 62, no. 10, pp. 132–138, 2024.

[13] H. Li, M. Nerini, S. Shen, and B. Clerckx, "A tutorial on beyond-diagonal reconfigurable intelligent surfaces: Modeling, architectures, system design and optimization, and applications," 2025. [Online]. Available: https://arxiv.org/abs/2505.16504

[14] X. Xie, C. He, X. Ma, F. Gao, Z. Han, and Z. J. Wang, "Joint precoding for active intelligent transmitting surface empowered outdoor-to-indoor communication in mmwave cellular networks," *IEEE Transactions on Wireless Communications*, vol. 22, no. 10, pp. 7072–7086, 2023.

[15] S. Shen, B. Clerckx, and R. Murch, "Modeling and architecture design of reconfigurable intelligent surfaces using scattering parameter network analysis," *IEEE Transactions on Wireless Communications*, vol. 21, no. 2, pp. 1229–1243, 2022.

[16] M. Nerini, S. Shen, and B. Clerckx, "Closed-form global optimization of beyond diagonal reconfigurable intelligent surfaces," *IEEE Transactions on Wireless Communications*, vol. 23, no. 2, pp. 1037–1051, 2024.

[17] H. Li, S. Shen, and B. Clerckx, "Beyond diagonal reconfigurable intelligent surfaces: From transmitting and reflecting modes to single-, group-, and fully-connected architectures," *IEEE Transactions on Wireless Communications*, vol. 22, no. 4, pp. 2311–2324, 2023.

[18] M. Nerini, S. Shen, H. Li, and B. Clerckx, "Beyond diagonal reconfigurable intelligent surfaces utilizing graph theory: Modeling, architecture design, and optimization," *IEEE Transactions on Wireless Communications*, vol. 23, no. 8, pp. 9972–9985, 2024.

[19] Z. Wu and B. Clerckx, "Beyond diagonal ris in multiuser mimo: Graph theoretic modeling and optimal architectures with low complexity," 2025. [Online]. Available: https://arxiv.org/abs/2502.16509

[20] X. Zhou, T. Fang, and Y. Mao, "A novel q-stem connected architecture for beyond-diagonal reconfigurable intelligent surfaces," 2024. [Online]. Available: https://arxiv.org/abs/2411.18480

[21] Z. Liu, X. Chu, D. López-Pérez, and N. Tang, "Deployment strategy of intelligent omni-surface-assisted outdoor-to-indoor millimeter-wave communications," *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, pp. 19 172–19 182, 2024.

[22] J. Zhang and D. M. Blough, "Optimizing coverage with intelligent surfaces for indoor mmwave networks," in *IEEE INFOCOM 2022 - IEEE Conference on Computer Communications*, 2022, pp. 830–839.

[23] R. Liang, J. Fan, H. Liu, Y. Ge, and J. Zhang, "Dual-hop hybrid irs-aided outdoor-to-indoor mmwave communications," *IEEE Communications Letters*, vol. 26, no. 12, pp. 2979–2983, 2022.

[24] M. Nerini, S. Shen, H. Li, M. Di Renzo, and B. Clerckx, "A universal framework for multiport network analysis of reconfigurable intelligent surfaces," *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 14 575–14 590, 2024.

[25] H. Li, M. Nerini, S. Shen, and B. Clerckx, "Beyond diagonal reconfigurable intelligent surfaces in wideband ofdm communications: Circuit-based modeling and optimization," *IEEE Transactions on Wireless Communications*, pp. 1–1, 2025.

[26] H. Li and B. Clerckx, "Non-reciprocal beyond diagonal ris: Multiport network models and performance benefits in full-duplex systems," 2024. [Online]. Available: https://arxiv.org/abs/2411.04370

[27] A. Mahmood, T. X. Vu, W. U. Khan, S. Chatzinotas, and B. Ottersten, "Joint computation and communication resource optimization for beyond diagonal uav-irs empowered mec networks," 2024. [Online]. Available: https://arxiv.org/abs/2311.07199

[28] A. Mishra, Y. Mao, C. D'Andrea, S. Buzzi, and B. Clerckx, "Transmitter side beyond-diagonal reconfigurable intelligent surface for massive mimo networks," *IEEE Wireless Communications Letters*, vol. 13, no. 2, pp. 352–356, 2024.

[29] K. Chen and Y. Mao, "Transmitter side beyond-diagonal ris for mmwave integrated sensing and communications," in *2024 IEEE 25th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2024, pp. 951–955.

[30] H. Li, S. Shen, M. Nerini, and B. Clerckx, "Reconfigurable intelligent surfaces 2.0: Beyond diagonal phase shift matrices," *IEEE Communications Magazine*, vol. 62, no. 3, pp. 102–108, 2024.

[31] D. Kumar, J. Kaleva, and A. Tölli, "Blockage-aware reliable mmwave access via coordinated multi-point connectivity," *IEEE Transactions on Wireless Communications*, vol. 20, no. 7, pp. 4238–4252, 2021.

[32] T. Bai, R. Vaze, and R. W. Heath, "Analysis of blockage effects on urban cellular networks," *IEEE Transactions on Wireless Communications*, vol. 13, no. 9, pp. 5070–5083, 2014.

[33] C. Slezak, V. Semkin, S. Andreev, Y. Koucheryavy, and S. Rangan, "Empirical effects of dynamic human-body blockage in 60 ghz communications," *IEEE Communications Magazine*, vol. 56, no. 12, pp. 60–66, 2018.

[34] S. S. Christensen, R. Agarwal, E. De Carvalho, and J. M. Cioffi, "Weighted sum-rate maximization using weighted mmse for mimo-bc beamforming design," *IEEE Transactions on Wireless Communications*, vol. 7, no. 12, pp. 4792–4799, 2008.

[35] M. Grant and S. Boyd, "CVX: Matlab software for disciplined convex programming, version 2.1," https://cvxr.com/cvx, Mar. 2014.

[36] L. Zhang, R. Zhang, Y.-C. Liang, Y. Xin, and H. V. Poor, "On gaussian mimo bc-mac duality with multiple transmit covariance constraints," *IEEE Transactions on Information Theory*, vol. 58, no. 4, pp. 2064–2078, 2012.

[37] K. Shen and W. Yu, "Fractional programming for communication systems—part ii: Uplink scheduling via matching," *IEEE Transactions on Signal Processing*, vol. 66, no. 10, pp. 2631–2644, 2018.

[38] ——, "Fractional programming for communication systems—Part I: Power control and beamforming," *IEEE Transactions on Signal Processing*, vol. 66, no. 10, pp. 2617–2630, 2018.

[39] P.-A. Absil, R. Mahony, and R. Sepulchre, "Optimization algorithms on matrix manifolds," in *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2009.

[40] T. Fang and Y. Mao, "A low-complexity beamforming design for beyond-diagonal ris aided multi-user networks," *IEEE Communications Letters*, vol. 28, no. 1, pp. 203–207, 2024.

[41] Y. Zhao, H. Li, B. Clerckx, and M. Franceschetti, "Mimo channel shaping and rate maximization using beyond diagonal ris," 2025. [Online]. Available: https://arxiv.org/abs/2407.15196

[42] T. Fang, Y. Mao, S. Shen, Z. Zhu, and B. Clerckx, "Fully connected reconfigurable intelligent surface aided rate-splitting multiple access for multi-user multi-antenna transmission," in *2022 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2022, pp. 675–680.

[43] C. Oestges and B. Clerckx, *MIMO wireless communications: from real-world propagation to space-time code design*. Academic Press, 2010.

[44] C. Yuen and B. M. Hochwald, "Achieving near-capacity at low snr on a multiple-antenna multiple-user channel," *IEEE Transactions on Communications*, vol. 57, no. 1, pp. 69–74, 2009.

[45] G. Yang, J. Du, and M. Xiao, "Maximum throughput path selection with random blockage for indoor 60 ghz relay networks," *IEEE Transactions on Communications*, vol. 63, no. 10, pp. 3511–3524, 2015.